

Functional Analysis

Labix

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Abstract

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1 The Topology on Normed Spaces

1.1 Norms and Metrics

Definition 1.1.1 (Norms)

Let $\mathbb{F}$ be a field. Let V be a vector space over $\mathbb{F}$. A norm on V is a function $\|\cdot\| : V \rightarrow \mathbb{R}$ such that the following are true.

- Non-negativity: For all $x \in V$, we have $\|x\| \geq 0$.
- Positive-Definite: For all $x \in V$, we have $\|x\| = 0$ if and only if $x = 0$.
- Absolute homogeneity: For all $\lambda \in \mathbb{R}$ and all $x \in V$, we have $\|\lambda x\| = |\lambda| \|x\|$.
- For all $x, y \in V$, we have $\|x + y\| \leq \|x\| + \|y\|$.

In this case we say that $(V, \|\cdot\|)$ is a normed vector space over $\mathbb{R}$.

Theorem 1.1.2 (The Parallelogram Law over $\mathbb{R}$)

Let $(V, \|\cdot\|)$ be a normed vector space over $\mathbb{R}$. Then there exists an inner product $\langle -, - \rangle$ such that $\|v\| = \sqrt{\langle v, v \rangle}$ for all $v \in V$ if and only if

$$2\|x\|^2 + 2\|y\|^2 = \|x + y\|^2 + \|x - y\|^2$$

for all $x, y \in V$.

Theorem 1.1.3 (The Parallelogram Law over $\mathbb{C}$)

Let $(V, \|\cdot\|)$ be a normed vector space over $\mathbb{C}$. Then there exists a Hermitian inner product $\langle -, - \rangle$ such that $\|v\| = \sqrt{\langle v, v \rangle}$ for all $v \in V$ if and only if

$$2\|x\|^2 + 2\|y\|^2 = \|x + y\|^2 + \|x - y\|^2$$

for all $x, y \in V$.

Definition 1.1.4 (The Induced Metric from Norms)

Let $(V, \|\cdot\|)$ be a normed vector space over $\mathbb{R}$. Define the induced metric to be the function $d : V \times V \rightarrow \mathbb{R}$ given by

$$d(x, y) = \|x - y\|$$

Definition 1.1.5 (Equivalent Norms)

Let k be a field. Let V be a vector space over k . Let $\|\cdot\|_1$ and $\|\cdot\|_2$ be two norms on V . We say that the two norms are equivalent if there exists $c_1, c_2 \in k$ such that

$$c_1 \|x\|_1 \leq \|x\|_2 \leq c_2 \|x\|_1$$

for all $x \in V$.

Proposition 1.1.6

Let k be a field. Let V be a vector space over k . Let $\|\cdot\|_1$ and $\|\cdot\|_2$ be two norms on V . Then $\|\cdot\|_1$ and $\|\cdot\|_2$ are equivalent if and only if the induced metric topologies are equal.

Proposition 1.1.7

Let k be a field. Let V be a vector space over k . Then all norms on V are equivalent.

1.2 Riesz's Lemma

Proposition 1.2.1

Let k be a field. Let V be a finite dimensional normed space. Let $U \subseteq V$ be a subset. Then U is compact if and only if U is closed and bounded.

Lemma 1.2.2 (Riesz's Lemma) Let $(X, \|\cdot\|)$ be a normed space and Y a non-empty closed subspace of X not equal to X . Then there exists $x \in X$ with $\|x\| = 1$ such that $\|x - y\| \geq \frac{1}{2}$ for every $y \in Y$.

Proposition 1.2.3 Let V be a normed space. Let $U \subseteq V$ be a finite dimensional vector subspace of V . Then U is closed.

Proposition 1.2.4 Let V be a normed space. Then V is finite dimensional if and only if $B_1(0)$ is compact.

1.3 Isomorphisms that Preserve Distance

Definition 1.3.1 (Isometrically Isomorphic Normed Spaces) Two normed spaces V and W are isometrically isomorphic if there is a surjective linear isometry $L : V \rightarrow W$. In this case, we write $V \cong W$.

Proposition 1.3.2 Let $(X, \|\cdot\|_X)$ be a normed space. Let V be a vector space over the same field and $L : V \rightarrow X$ a linear isomorphism. Then the pullback norm

$$\|v\|_V = \|L(v)\|_X$$

defines a norm on V . In particular, $L : (V, \|\cdot\|_V) \rightarrow (X, \|\cdot\|_X)$ is a linear isometry.

2 Vector Spaces with a Topological Structure

2.1 Topological Vector Spaces

Definition 2.1.1 (Topological Field)

Let k be a field equipped with a topology. We say that k is a topological field if the following are true.

- The addition map $+: k \times k \rightarrow k$ is continuous.
- The multiplication map $\cdot: k \times k \rightarrow k$ is continuous.
- The inverse map $(\cdot)^{-1}: k^\times \rightarrow k^\times$ is continuous.

Definition 2.1.2 (Topological Vector Space)

Let k be a topological field. Let V be a vector space over a field k that is also a topological space. We say that V is a topological vector space if the following are true.

- The addition map $+: V \times V \rightarrow V$ is continuous.
- The scalar multiplication map $\cdot: k \times V \rightarrow V$ is continuous.

Lemma 2.1.3

Let k be a topological field. Let V be a normed space over k . Then the metric topology induced by the induced metric from the norm equips V with the structure of a topological vector space.

Let k be a topological field. Let V be a vector space. Then any norm on V makes V a topological vector space. Moreover, by the above proposition, there is a uniquely defined topology on V as long as at least one norm on V exists.

2.2 The Topology of Normed Spaces over $\mathbb{R}$

Proposition 2.2.1

Let V be a finite dimensional normed space over $\mathbb{R}$. Then the induced metric on V is complete.

Proposition 2.2.2

Let V, W be finite dimensional normed spaces over $\mathbb{R}$. Let $T: V \rightarrow W$ be a linear map. Then T is continuous.

3 Banach Spaces

3.1 Banach Spaces

Definition 3.1.1 (Banach Spaces) Let $(V, \|\cdot\|)$ be a normed space. We say that V is a Banach space if V is complete as a metric space.

Proposition 3.1.2 Let V be a finite dimensional normed space. Then V is a Banach space.

Lemma 3.1.3 Suppose that $(X, \|\cdot\|_X) \cong (Y, \|\cdot\|_Y)$ are isometrically isomorphic. Then X is a Banach space if and only if Y is a Banach space.

Lemma 3.1.4 If $\|\cdot\|_1$ and $\|\cdot\|_2$ are equivalent norms on a vector space X then $(X, \|\cdot\|_1)$ is complete if and only if $(X, \|\cdot\|_2)$ is complete.

Proposition 3.1.5 Both $\mathbb{R}^n$ and $\mathbb{C}^n$ are complete.

3.2 Separability of Banach Spaces

Recall the notion of separability: A space is separable if it has a countably dense subset.

Lemma 3.2.1 Let X be a normed space. Then the following are equivalent.

- X is separable
- The set $\{x \in X \mid \|x\| = 1\}$ is separable
- X contains a sequence $(x_n)_{n \in \mathbb{N}}$ whose linear span is dense.

3.3 The Completion of a Normed Space

4 The Space of Bounded Linear Maps

4.1 Basic Definitions

Definition 4.1.1 (Bounded Linear Maps) Let V, W be normed space over a field k . Let $T : V \rightarrow W$ be a linear map. We say that T is bounded if there exists $M > 0$ such that

$$\|T(x)\|_W \leq M\|x\|_V$$

for all $x \in V$.

Lemma 4.1.2 Let V, W be normed spaces over a field k . Let $T : V \rightarrow W$ be a linear map. Then T is bounded if and only if T is continuous.

Definition 4.1.3 (The Space of Bounded Linear Maps) Let V, W be normed spaces over a field k . Define the space of bounded linear maps to be the vector subspace

$$B(V, W) = \{T \in \text{Hom}_k(V, W) \mid T \text{ is bounded} \}$$

together with the operator norm $\|\cdot\|_{\text{op}} : B(V, W) \rightarrow k$ defined by

$$\|T\|_{\text{op}} = \sup\{\|T(v)\|_W \mid v \in V \text{ such that } \|v\|_V = 1\}$$

Lemma 4.1.4 Let V, W be normed spaces. If V is finite dimensional, then we have

$$B(V, W) = \text{Hom}_k(V, W)$$

Lemma 4.1.5 Let V, W be normed spaces over a field k . Then $B(V, W)$ is a Banach space.

4.2 Invertibility

Corollary 4.2.1 Let $T \in B(X, Y)$ for X, Y vector spaces. Then $\ker(T)$ is a closed linear subspace of X .

Definition 4.2.2 (Bounded Invertible) A linear map $T \in B(X, Y)$ is bounded invertible if there exists $S \in B(X, Y)$ such that $S \circ T$ and $T \circ S$ are the identity.

Lemma 4.2.3 Suppose that X and Y are normed spaces. Then for any $T \in B(X, Y)$ the following are equivalent.

- T is bounded invertible
- T is a bijection and $T^{-1} \in B(X, Y)$
- T is surjective and for some $c > 0$, $\|T(x)\|_Y \geq c\|x\|_X$ for every $x \in X$.

Corollary 4.2.4 If X is finite dimensional then a linear operator $T : X \rightarrow X$ is invertible if and only if $\ker(T) = \{0\}$.

4.3 The Hahn Banach Theorem

Definition 4.3.1 (Continuous Dual Space) Let X be a normed space. Denote the continuous dual space of X to be the subspace

$$X' = B(X, K)$$

of the dual space X^* .

Notice that since continuity is the same as boundedness, this notation of $B(X, K)$ coincides with the set of all bounded linear operators.

Lemma 4.3.2 Let X be a Banach space. Then X' is also a Banach space.

Proof Since we know that $B(X, Y)$ is a Banach space for any normed space X and Banach space Y , choose $Y = \mathbb{R}$ and we are done. ■

Theorem 4.3.3 (The Real Hahn-Banach Theorem) Let X be a real vector space. Let p be a convex function on X . Let X_0 be a linear subspace of X and let f be a linear functional on Y satisfying

$$f(x) \leq p(x)$$

for all $x \in Y$. Then f can be extended to a linear functional F on all of X satisfying the condition

$$F(x) \leq p(x)$$

for all $x \in X$ and $F|_Y = f$.

5 Hilbert Spaces

5.1 Hilbert Spaces

Definition 5.1.1 (Hilbert Spaces) Let V be an inner product space. We say that V is a Hilbert space if V is a Banach space.

5.2 Orthogonality in Hilbert Spaces

Definition 5.2.1 (Orthonormal Set) A subset E of a Hilbert space is orthonormal if $\|e\| = 1$ for all $e \in E$ and $\langle e_1, e_2 \rangle = 0$ for all $e_1, e_2 \in E$ with $e_1 \neq e_2$.

Lemma 5.2.2 (Bessel's Inequality) Let V be an inner product space and $(e_n)_{n \in \mathbb{N}}$ be an orthonormal sequence. Then for any $x \in V$ we have

$$\sum_{k=1}^{\infty} |\langle x, e_k \rangle|^2 \leq \|x\|^2$$

Lemma 5.2.3 Let H be a Hilbert space and $(e_n)_{n \in \mathbb{N}}$ an orthonormal sequence in H . Then the series $\sum_{k=1}^{\infty} a_k e_k$ converges if and only if

$$\sum_{k=1}^{\infty} |a_k|^2 < \infty$$

and

$$\left\| \sum_{k=1}^{\infty} a_k e_k \right\|^2 = \sum_{k=1}^{\infty} |a_k|^2$$

Corollary 5.2.4 Let H be a Hilbert space and $(e_n)_{n \in \mathbb{N}}$ an orthonormal sequence in H . Then for any x the sequence

$$\sum_{k=1}^{\infty} \langle x, e_k \rangle e_k$$

converges.

Proposition 5.2.5 Let $E = (e_n)_{n \in \mathbb{N}}$ be an orthonormal sequence in a Hilbert space H . Then the following are equivalent.

- E is a basis for H
- For any $x \in H$ we have $x = \sum_{k=1}^{\infty} \langle x, e_k \rangle e_k$
- $\|x\|^2 = \sum_{k=1}^{\infty} |\langle x, e_k \rangle|^2$ for all $x \in H$
- $\langle x, e_n \rangle = 0$ for all n implies $x = 0$
- $\text{span}(E) = H$

5.3 Separability of Hilbert Spaces

Proposition 5.3.1 An infinite-dimensional Hilbert space is separable if and only if it has a countable orthonormal basis.

Theorem 5.3.2 Any infinite dimensional separable Hilbert space H is isometric to $l^2(K)$ for some field K .

5.4 The Adjoint of Maps of Hilbert Spaces

Theorem 5.4.1 (Riesz Representation Theorem)

Theorem 5.4.2 Let H, K be Hilbert spaces. Let $A : H \rightarrow K$ be a bounded linear map. Then there exists a unique bounded linear map $A^* : K \rightarrow H$ such that

$$\langle A(x), y \rangle = \langle x, A^*(y) \rangle$$

for all $x \in H$ and $y \in K$.

6 Spectral Theory

6.1 Spectrum

Definition 6.1.1 (Resolvent and Spectrum) Let X be a complex Banach space and $T \in B(X)$. Define the resolvent set of T to be

$$\rho(T) = \{\lambda \in \mathbb{C} \mid T - \lambda I \text{ is invertible}\}$$

Define the spectrum of T to be

$$\sigma(T) = \mathbb{C} \setminus \rho(T) = \{\lambda \in \mathbb{C} \mid T - \lambda I \text{ is not invertible}\}$$

Notice that in the case that X is finite dimensional, the resolvent set of any linear operator is just its set of eigenvalues.

Lemma 6.1.2 Suppose that $T \in B(X)$ and that $(\lambda_n)_{n \in \mathbb{N}}$ are distinct eigenvalues of T . Then any set $(e_n)_{n \in \mathbb{N}}$ of corresponding eigenvectors is linearly independent.

Theorem 6.1.3 Let X be a Banach space and $T \in B(X)$ such that $T^{-1} \in B(X)$. Then for any $U \in B(X)$, with $\|U\| < \|T^{-1}\|^{-1}$, we have

$$\|(T + U)^{-1}\| \leq \frac{\|T^{-1}\|}{1 - \|U\|\|T^{-1}\|}$$

Lemma 6.1.4 Let X be a Banach space. Let $T \in B(X)$. Then $\sigma(T)$ is a closed subset of $\{\lambda \in \mathbb{C} \mid |\lambda| \leq \|T\|\}$.